Chapter 6: Anatomy of Flowering Plants

Monocot Leaf

Isobilateral leaf | **Stomata roughly equal on both surfaces** | **Mesophyll NOT differentiated into palisade/spongy** | **Bulliform cells on adaxial epidermis control rolling** | **Vascular bundle conjoint & closed, endarch**

Epidermis

A monocot (isobilateral) leaf also has adaxial and abaxial epidermis, both covered by a conspicuous cuticle, but here **stomata are present on both surfaces** in roughly equal numbers.

Bulliform Cells

In grasses, certain adaxial epidermal cells along the veins modify into large, empty, colourless cells called **bulliform cells**. The leaf surface is exposed when the bulliform cells absorb water and become **turgid**; when the bulliform cells lose water they become **flaccid**, and the leaf curls/rolls inward — minimizing water loss.

Mesophyll

The tissue between the upper and lower epidermis is the mesophyll, made up of chlorophyll-containing parenchyma cells which, unlike the dicot leaf, are **not differentiated** into palisade and spongy parenchyma.

Vascular Bundle

Surrounded by a layer of parenchymatous **bundle sheath cells**. The vascular bundle is **conjoint and closed**, with **endarch** xylem.

Monocot Leaf (Isobilateral) – Key Regions

Region	Key Feature
Epidermis	Adaxial + abaxial, stomata on both surfaces, bulliform cells
Mesophyll	Not differentiated (no palisade/spongy split)
Vascular bundle	Bundle sheath, conjoint, closed, endarch

Isobilateral Leaf, Signature Features

- Stomata equal on both surfaces
- Bulliform cells (leaf rolling)
- Undifferentiated mesophyll

- Parallel venation